

ANIL PRADHAN

Full Stack Developer · AI/Agent Systems · Backend Engineering

+91 8917476908 · anilpradhan9644@gmail.com · linkedin.com/in/anil-pradhan543 · github.com/Anil-Pradhan-web
[Portfolio](#) · [LeetCode](#) · [GeeksForGeeks](#)

EDUCATION

B.Tech – CSE | CGPA: 9.11/10 · ITER, SOA University, Bhubaneswar 2024 – 2028

Class XII – PCM with IT | 85% · Odisha Board of Secondary Education 2022 – 2024

EXPERIENCE

Campus Ambassador · E-Cell, IIT Kanpur · Remote Dec 2025 – Jan 2026

- ▶ Promoting IIT Kanpur's E-Summit Events at ITER, SOA University via social media, campus networking, and community-building activities.

PROJECTS

AI Career Mentor Feb 2026 – Present

Live: [ai-career-mentor-anil.vercel.app](#)

Next.js 14 · FastAPI · LangGraph · ChromaDB · PostgreSQL · Upstash Redis · Groq · NVIDIA NIM · Gemini 2.5 Flash · Docker · GitHub Actions

- ▶ **AI Orchestration, RAG & Resilience:** Architected a hybrid AI career intelligence platform using LangGraph, designing a parallel DAG to orchestrate 4 specialized agents concurrently (Resume, Market, LinkedIn, Roadmap) reducing latency by 60%; integrated a semantic ChromaDB RAG pipeline with ONNX embeddings and a unified Agent Registry featuring custom Circuit Breakers (300s cooldown) and automatic fallback routing across NVIDIA NIM, Groq, and Gemini guaranteeing 100% API uptime.
- ▶ **Real-Time Communication & Streaming:** Developed full-duplex voice assistant proxying Gemini Multimodal Live API via secure WebSockets (relaying 16kHz/24kHz mono PCM audio), a 7-phase FSM mock interview console with live Monaco Editor sync and look-ahead Edge-TTS synthesis, and FastAPI SSE to stream async LangGraph node execution logs to the frontend.
- ▶ **Observability, Security & Data Pipelines:** Built a high-throughput observability pipeline tracking live metrics in Redis keyspace (active sessions, provider latencies) rolling up daily into Neon PostgreSQL, hardened by a global Loguru error interceptor, multi-layer SlowAPI rate limits, 48-hour feature locks, and a 4-layer sanitization engine preventing prompt injection attacks.
- ▶ **DevOps & CI/CD:** Docker Compose with multi-stage Dockerfiles (2-stage backend, 3-stage Next.js frontend) cutting image sizes by 50%; GitHub Actions executing parallel frontend builds, 106 Pytest tests, pip-audit security scans, and Newman automated API integration testing.
- ▶ **Frontend Performance:** Next.js 14 SPA with Lenis Smooth Scroll for lag-free inertial navigation, glassmorphic MobileBlocker viewport interceptor preserving UI fidelity above 1024px, and custom Axios interceptors automating JWT header attachment, silent token refresh, and rate-limit handling.

VolunteerIQ Mar 2026 – Apr 2026

Live: [volunteer-ig-clutchcode.vercel.app](#)

Next.js 14 · FastAPI · PostgreSQL · Gemini 2.5 Flash · Groq · Firebase Auth · Mapbox GL · Docker

- ▶ Developed backend APIs and AI services for an NGO coordination platform, enabling automated survey analysis, task generation, volunteer matching, and contextual AI-powered assistance.
- ▶ Built multi-format document processing pipelines (PDF, CSV, DOCX, TXT) integrated with Gemini AI and Groq fallback mechanisms to ensure reliable humanitarian intelligence workflows.
- ▶ Designed secure backend infrastructure using FastAPI, PostgreSQL, SQLAlchemy, Firebase authentication, Mapbox GL live ops map, Docker containerization, and REST APIs supporting real-time NGO operations.

TECHNICAL SKILLS

Languages: Java · Python · SQL · JavaScript · TypeScript

Frontend: HTML5 · CSS3 · Tailwind CSS · shadcn/ui · React 18 · Next.js 14 · Recharts

Backend: FastAPI · Uvicorn · LangGraph · SQLAlchemy 2.0 · Alembic · Pydantic · pdfplumber · Pytest · edge-tts · SlowAPI · WebSocket · sentry-sdk · prometheus-fastapi-instrumentator

AI / LLM & RAG: Groq (Llama 3.3 70B) · NVIDIA NIM · Gemini 2.5 Flash · Gemini Live API · ChromaDB · Serper.dev · Tavily AI · DuckDuckGo

Databases: Neon Postgres · SQLite · Upstash Redis · ChromaDB (Vector Store)

Auth & Security: Google OAuth 2.0 · Firebase Authentication · JWT (python-jose) · bcrypt

DevOps & Infra: Docker · GitHub Actions (CI/CD) · Vercel · Render.com · GHCR

Tools: VS Code · Git · Swagger UI · Postman · Cline · GitHub Copilot · Gemini Code Assist · OpenAI Codex · AntiGravity IDE

ACHIEVEMENTS & CERTIFICATIONS

- ▶ Google Cloud Arcade Program (2024–25): Completed with 75 Arcade Points across hands-on GCP labs and challenges.
- ▶ GeeksForGeeks 80 Days Coding Challenge: 200+ problems solved across GFG and LeetCode (Arrays, Strings, Recursion, DP, Graphs). Certificate + T-shirt awarded.
- ▶ Active in developer communities including GDG and Google tech events, continuously exploring AI systems, cloud infrastructure, backend engineering, and networking through offline meetups and bootcamps.
- ▶ Participated in multiple IIT/NIT coding competitions, hackathons, and developer events focused on AI systems, backend engineering, and problem solving.
- ▶ Built production-grade AI products across Microsoft AI Dev Days Hackathon (\$80K+), Amazon Nova AI Hackathon (\$95K+), and Google Solution Challenge 2026, transforming ideas into deployable full-stack applications under hackathon timelines.